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DIRICHLET AVERAGES OF $x^t \log x$*

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Abstract. A neglected class of special functions may be described as Dirichlet averages of $x^t \log x$ or equivalently as derivatives of hypergeometric R-functions with respect to the degree of homogeneity. Special cases include the derivative of a Legendre function with respect to the degree and the derivative of Gauss's hypergeometric function with respect to a numerator parameter. There are connections with the logarithmic derivative of the gamma function, with Euler's dilogarithm, and with ${}_3F_2$; some special cases of ${}_3F_2$ are thereby evaluated. Applications include a two-point boundary-value problem, mean values, series expansions of elliptic integrals, integral tables, and several physics problems. The discussion of various properties emphasizes series expansions, quadratic transformations, inequalities, and evaluation of special cases, including certain cases of the derivative of a Legendre function with respect to the degree.

Key words. Dirichlet averages, logarithms, hypergeometric functions, R-functions, elliptic integrals, Legendre functions

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1. Introduction. A class of special functions that seems not to have been discussed systematically arises in the asymptotic expansion of elliptic integrals and in various other problems. The prototype of this class is the derivative of a Legendre function P_ν with respect to the degree ν. The general case is the derivative of the multivariate hypergeometric function R_t with respect to the degree t of homogeneity. It can equivalently be thought of as a Dirichlet average of $\partial x^t/\partial t = x^t \log x$. After defining this average, denoted by L_t, we shall list some places where it occurs.

We recall first the definition of the R-function [3, (5.9-1)]. Let b and z be k-tuples with components in the open right-half complex plane, $\mathbb{C}_>$. The Dirichlet average of x^t, $t \in \mathbb{C}$, is defined for $k \geqq 2$ by

$$R_t(b, z) = \int (u \cdot z)^t \, d\mu_b(u), \tag{1.1}$$

where $u \cdot z = \sum_{i=1}^k u_i z_i$ is a convex combination of the variables $z_1, \cdots, z_k$ and μ_b is a Dirichlet measure (i.e. a multivariate beta distribution with possibly complex parameters $b_1, \cdots, b_k$). The integration extends over all nonnegative weights $u_1, \cdots, u_k$ whose sum is unity. If $k = 1$ we define $R_t(b, z) = z^t$. The analytic continuation [3, Sec. 6.8] of $R_t(b, z)/\Gamma(c)$, $c = \sum_{i=1}^k b_i$, is entire in t and b and holomorphic in z on $\mathbb{C}_0^k$, where $\mathbb{C}_0$ is the complex plane slit along the nonpositive real axis.

The subject of this paper is the function L_t defined by

$$L_t(b, z) = \frac{\partial}{\partial t} R_t(b, z). \tag{1.2}$$

If b, $z \in \mathbb{C}_>^k$, $k \geqq 2$, we see from (1.1) that

$$L_t(b, z) = \int (u \cdot z)^t \log (u \cdot z) \, d\mu_b(u). \tag{1.3}$$

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That is, L_t is a Dirichlet average of $x^t \log x$. The case $t=0$ is discussed briefly in [3, Ex. 5.9-12 and p. 305]. If $k=1$ we define $L_t(b, z)=z^t \log z$. We shall often display the components of b and z by writing $L_t(b_1, \cdots, b_k; z_1, \cdots, z_k)$. If $k=2$, $b=(\beta, \beta')$ and $z=(x, y)$, (1.3) becomes

$$L_t(\beta, \beta'; x, y)=\frac{\Gamma(\beta+\beta')}{\Gamma(\beta)\Gamma(\beta')}\int_0^1 [ux+(1-u)y]^t \log[ux+(1-u)y] \cdot u^{\beta-1}(1-u)^{\beta'-1}\,du, \tag{1.4}$$

where β, β', x, y have positive real parts and t is unrestricted. The case in which x or y is 0 will be discussed in § 4.

We list some examples to show that L_t is worth studying.

Example 1. The difference between two values of the logarithmic derivative of the gamma function is

$$\psi(\beta)-\psi(\gamma)=L_0(\beta, \gamma-\beta; 1, 0), \quad \operatorname{Re}\beta>0, \quad \gamma\neq 0, -1, -2, \cdots. \tag{1.5}$$

Example 2. The derivative of Gauss' hypergeometric function with respect to a numerator parameter is

$$\frac{\partial}{\partial\alpha}\,{}_2F_1(\alpha, \beta; \gamma; z)=-L_{-\alpha}(\beta, \gamma-\beta; 1-z, 1). \tag{1.6}$$

If α is a negative integer, the ${}_2F_1$-function is a polynomial; if α is close to a negative integer, the function can be approximated to first order with the help of the derivative. Equation (1.6) can easily be generalized from ${}_2F_1$ to Appell's F_1 or Lauricella's F_D by expressing these functions in terms of R_t (e.g. see [3, Ex. 6.3-5]).

Example 3. The derivative of an associated Legendre function with respect to its degree is

$$\frac{\partial}{\partial\nu} P_\nu^{-\mu}(\cos\theta)=\frac{\sin^\mu\theta}{2^\mu\Gamma(1+\mu)} L_{\nu-\mu}(\mu+1/2, \mu+1/2; e^{i\theta}, e^{-i\theta}). \tag{1.7}$$

This derivative is useful in antenna theory [13, pp. 114-115]. Special cases are evaluated in closed form in (6.14) and (6.19).

Example 4. For any real t and positive x, a, b, the solution of the two-point boundary-value problem $y''=x^t \log x$, $y(a)=y(b)=0$, is

$$y(x)=\tfrac{1}{2}(x-a)(x-b)L_t(1, 1, 1; x, a, b). \tag{1.8}$$

Closed formulas are given in (8.10), (8.12), and (8.14).

Example 5. Let $x_1, \cdots, x_k$ be positive numbers and $w_1, \cdots, w_k$ positive weights with $\sum w_i=1$. Then $\exp L_0(cw_1, \cdots, cw_k; x_1, \cdots, x_k)$, $c>0$, is a homogeneous mean value of the x_i that increases strictly with c unless the x_i are all equal. It tends to the weighted geometric mean $\prod x_i^{w_i}$ as $c\to 0+$ and the weighted arithmetic mean $\sum w_ix_i$ as $c\to+\infty$.

Example 6. Series expansions of elliptic integrals and some other R-functions [6], [5] near their singularities involve L_t. For example [6, (1.20)],

$$\begin{aligned}(zw)^{1/2}\int_0^\infty [(t+x)(t+y)(t+z)(t+w)]^{-1/2}\,dt &= \sum_{n=0}^{\infty} [-L_n(1/2, 1/2; x, y)R_n(1/2, 1/2; z^{-1}, w^{-1}) \\ &\quad - R_n(1/2, 1/2; x, y)L_n(1/2, 1/2; z^{-1}, w^{-1})]\end{aligned} \tag{1.9}$$

where x and y are nonnegative, z and w are positive, and $0 < \max\{x, y\}/\min\{z, w\} < 1$. Convergence is fast near the logarithmic singularity at $x = y = 0$. A closed formula for $L_n(1/2, 1/2; x, y)$, which was not known when [6] was written, is given in (6.13).

Example 7. Integrals with a logarithm in the integrand are met in various applied problems and can sometimes be put in the form (1.4). Two such integrals occurring in diffusion problems associated with crystal growth [14, (17), (36)] [11, (A4)] are constant multiples of $L_0(1/2, 1/2+1/n; z^2-1, z^2)$ and $L_0(1+a, 1-a; z-1, z+1)$, where n is a positive integer and a is an angle in units of π. In connection with the first example, Seeger [14, p. 6] remarks that he "did not succeed in expressing [the concentration] in terms of well-investigated functions." A more complicated integral (not found in tables) containing a logarithm and a Bessel function is

$$\begin{aligned}\int_0^\infty e^{-pt}\log(t)t^\nu J_\nu(xt)\,dt &= \pi^{-1/2}\Gamma(\nu+1/2)(2x)^\nu(p^2+x^2)^{-\nu-1/2}\\ &\quad\cdot[\psi(2\nu+1)-\log(p^2+x^2)+\tfrac{1}{2}L_0(\nu+1/2, 1/2; p^2, p^2+x^2)],\end{aligned} \tag{1.10}$$

where p, x, and $\nu+1/2$ are positive. An example with an integrand not containing a logarithm is

$$\int_0^1 u^{-1}(1-u)^{a-1}[{}_2F_1(a, b; c; ux)-1]\,du = -L_0(b, c-b; 1-x, 1), \tag{1.11}$$

where $\operatorname{Re} a > 0$, $|\arg(1-x)| < \pi$, and $c \neq 0, -1, -2, \cdots$. The case $a = b = \frac{1}{2}$, $c = 1$ occurs in the theory of neutron stars [8, Appendix B].

Equation (1.5) is a special case of (4.4). Equations (1.6) and (1.7) follow, respectively, from differentiating [3, (5.9-12) and (6.8-19)]. It can be verified that (1.8) satisfies the differential equation by using [3, (5.6-5), (5.6-10), (6.3-3)] to prove a more general result for $y'' = f(x)$. Example 5 comes from Theorem 7.2 or from [1, (2.6) and Thm. 1] and [7, Thm. 4]. Equation (1.9) is taken from [6, (1.18), (3.17)]. Equation (1.10) is proved by differentiating [3, Ex. 5.10-3] with respect to λ and using [3, (5.9-23)] and Equations (2.11), (6.4), and (2.3) of this paper. Equation (1.11) is derived by integrating the ${}_2F_1$-series term by term and comparing with (5.13).

This paper does not discuss higher derivatives of R_t with respect to t nor double Dirichlet averages of $x^t \log x$, although the latter occur in some problems such as the two-dimensional Ising model.

2. General properties. Several important properties of L_t follow immediately from the theory of Dirichlet averages or from properties of R_t:

(2.1) $L_t(b, z)/\Gamma(c)$, $c = \sum_{i=1}^k b_i$, is entire in t and b and holomorphic in z on $\mathbb{C}_0^k$. See (1.2).

(2.2) $L_t(b_1, \cdots, b_k; z_1, \cdots, z_k)$ is symmetric in the indices $1, \cdots, k$ [3, Thm. 5.2-3].

(2.3) A vanishing parameter b_i can be omitted along with the corresponding variable z_i [3, (6.3-3)].

(2.4) Equal variables can be replaced by a single variable if the corresponding parameters are replaced by their sum [3, (5.2.-3)].

In particular, if all variables are equal, then

$$L_t(b_1, \cdots, b_k; x, \cdots, x) = L_t(c, x) = x^t \log x.$$

From (1.3) we obtain

$$\begin{aligned} L_t(b, \lambda z) &= \lambda^t L_t(b, z) + \lambda^t R_t(b, z) \log \lambda \\ &= \lambda^t L_t(b, z) + R_t(b, \lambda z) \log \lambda, \\ L_0(b, \lambda z) &= L_0(b, z) + \log \lambda. \end{aligned} \tag{2.5}$$

Differentiation of [3, (6.8-15)] gives

$$L_t(b, z) = -\left(\prod_{i=1}^{k} z_i^{-b_i}\right) L_{-c-t}(b, z^{-1}), \tag{2.6}$$

where $c = \sum_{i=1}^{k} b_i \neq 0, -1, -2, \cdots$; $z \in \mathbb{C}_0^k$; and $z^{-1} = (z_1^{-1}, \cdots, z_k^{-1})$.

Since L_t is a Dirichlet average, it satisfies a system of Euler–Poisson equations [3, (5.4-2)],

$$[(z_i - z_j) D_i D_j + b_i D_j - b_j D_i] L_t(b, z) = 0, \qquad i, j = 1, 2, \cdots, k, \tag{2.7}$$

where $D_i = \partial/\partial z_i$. Differentiating [3, (5.9-2)] with respect to t, we find also the inhomogeneous differential equation

$$\sum_{i=1}^{k} z_i D_i L_t = t L_t + R_t \tag{2.8}$$

and the differential-difference equation

$$\sum_{i=1}^{k} D_i L_t = t L_{t-1} + R_{t-1}. \tag{2.9}$$

If we define $x + z = (x + z_1, \cdots, x + z_k)$, where z is independent of x, then (2.9) implies

$$\frac{d}{dx} L_t(b, x+z) = t L_{t-1}(b, x+z) + R_{t-1}(b, x+z). \tag{2.10}$$

Some useful relations are peculiar to the case of two variables:

$$L_t(\beta, \beta'; x, y) = \log (xy) R_t(\beta, \beta'; x, y) - x^{t+\beta'} y^{t+\beta} L_{-\beta-\beta'-t}(\beta', \beta; x, y), \tag{2.11}$$

where $x, y \in \mathbb{C}_0$ and $\beta + \beta' \neq 0, -1, -2, \cdots$. This follows from (2.6) and (2.5) or alternatively from [3, (5.9-21)].

$$\begin{aligned} \frac{\partial}{\partial t} R_{-a}(u+t, v-t; x, y) &= x^{-a} L_{t-v}(u+v-a, a; 1, y/x) \\ &= -y^{-a} L_{-u-t}(a, u+v-a; x/y, 1), \end{aligned} \tag{2.12}$$

where $x, y \in \mathbb{C}_0$ and $u + v \neq 0, -1, -2, \cdots$. On the left side use homogeneity to arrive at arguments 1 and y/x and then apply [3, (5.9-20)]. The second member equals the third by (2.6) and (2.2). We shall use (2.12) to derive (6.4) and (6.5).

In addition to the representation of L_t by the multiple integral (1.3), a representation by a single integral is obtained by differentiating the corresponding representation [3, (6.8-6), Ex. 6.8-8] of R_t:

$$\begin{aligned} & B(a, a')\{L_{-a}(b, z) + [\psi(a') - \psi(a)] R_{-a}(b, z)\} \\ &= \int_0^\infty t^{a'-1} \log (t) \prod_{i=1}^{k} (t + z_i)^{-b_i} \, dt \\ &= -\int_0^\infty t^{a-1} \log (t) \prod_{i=1}^{k} (1 + t z_i)^{-b_i} \, dt, \end{aligned} \tag{2.13}$$

where B is the beta function, $a+a'=\sum_{i=1}^k b_i$, a and a' have positive real parts, and $z\in\mathbb{C}_0^k$. For integrals with finite limits of integration, we find by differentiating [3, (8.1-2)] with respect to a that

$$\begin{aligned}\int_x^y (t-x)^{a-1}(y-t)^{a'-1}\log\left(\frac{y-t}{t-x}\right)\prod_{i=1}^k (z_i+w_it)^{-b_i}\,dt \\ = B(a,a')(y-x)^{a+a'-1}\prod_{i=1}^k (z_i+w_ix)^{-b_i} \\ \cdot\left\{L_{-a}\left(b,\frac{z+wy}{z+wx}\right)+[\psi(a')-\psi(a)]R_{-a}\left(b,\frac{z+wy}{z+wx}\right)\right\} \\ = B(a,a')(y-x)^{a+a'-1}\prod_{i=1}^k (z_i+w_iy)^{-b_i} \\ \cdot\left\{-L_{-a'}\left(b,\frac{z+wx}{z+wy}\right)+[\psi(a')-\psi(a)]R_{-a'}\left(b,\frac{z+wx}{z+wy}\right)\right\}\end{aligned} \tag{2.14}$$

where the conditions on the parameters are the same as for (2.13), $y>x$, $(z+wy)/(z+wx)$ denotes the k-tuple with ith component $(z_i+w_iy)/(z_i+w_ix)$, and $z_i+w_it\in\mathbb{C}_0$ for every $t\in[x,y]$ and every $i=1,\cdots,k$. The equality $a+a'=\sum_{i=1}^k b_i$ can always be satisfied by choice of b_k if we put $z_k=1$ and $w_k=0$.

An integral containing the confluent hypergeometric function S (the Dirichlet average of e^x) can be evaluated by differentiating [3, (5.10-11)] with respect to a:

$$\int_0^\infty t^{a-1}\log(t)S(b,-tz)\,dt=\Gamma'(a)R_{-a}(b,z)-\Gamma(a)L_{-a}(b,z), \tag{2.15}$$

where a and all components of z have positive real parts and $\sum_{i=1}^k b_i\neq 0,-1,-2,\cdots$. The functions of Kummer, Bessel and Whittaker are expressed in terms of S by [3, (5.8-7), (5.8-23), (5.12-20 to 27)].

3. Relations between associated functions. Two or more L-functions of k variables are said to be associated if the parameters $t, b_1,\cdots,b_k$ of each function differ by integers from the corresponding parameters of the other functions.

THEOREM 3.1. *Between any $k+1$ associated L-functions of k variables there exists a linear (possibly inhomogeneous) relation with coefficients that are polynomials in the variables and parameters. The inhomogeneous term, if any, is a linear combination of R-functions with polynomial coefficients.*

Proof. There is a linear homogeneous relation between any $k+1$ associated R-functions [3, Thm. 8.4-3] in which the coefficients are polynomials in the parameters as well as the variables. By (1.2), differentiation with respect to t produces a linear relation of the kind described between the corresponding L-functions.

Let $w_i=b_i/c$, $c=\sum_{i=1}^k b_i$, and let e_i be a k-tuple with unity in the ith place and zeros elsewhere. We list first some homogeneous relations:

$$L_t(b,z)=\sum_{i=1}^k w_iL_t(b+e_i,z), \tag{3.1}$$

$$L_{t+1}(b,z)=\sum_{i=1}^k w_iz_iL_t(b+e_i,z), \tag{3.2}$$

$$(z_i-z_j)L_t(b-e_h,z)+(z_j-z_h)L_t(b-e_i,z)+(z_h-z_i)L_t(b-e_j,z)=0. \tag{3.3}$$

The first and third are special cases of [3, (5.6-4), (5.6-11)], while the second comes from differentiating [3, (5.9-6)].

Differentiation of [3, (5.9-8), (5.9-9), (5.9-10)] yields, for $i=1,2,\cdots,k$,

$$(3.4)\quad (c-1)L_t(b-e_i,z)=(c+t-1)L_t(b,z)-tz_iL_{t-1}(b,z)+R_t(b,z)-z_iR_{t-1}(b,z),$$

$$(3.5)\quad D_iL_t(b,z)=w_itL_{t-1}(b+e_i,z)+w_iR_{t-1}(b+e_i,z),$$

$$(3.6)\quad (z_iD_i+b_i)L_t(b,z)=w_i(c+t)L_t(b+e_i,z)+w_iR_t(b+e_i,z).$$

From [3, Ex. 5.9-6] we see that

$$(3.7)\quad t(z_i-z_j)L_{t-1}(b,z)+(c-1)[L_t(b-e_i,z)-L_t(b-e_j,z)]+(z_i-z_j)R_{t-1}(b,z)=0,$$

$$(3.8)\quad \begin{aligned}&(c+t-1)(z_i-z_j)L_t(b,z)+(c-1)[z_jL_t(b-e_i,z)-z_iL_t(b-e_j,z)]\\&\quad+(z_i-z_j)R_t(b,z)=0.\end{aligned}$$

In the case of two variables, differentiation of [3, (5.9-24), (5.9-25)] gives

$$(3.9)\quad w_i(z_j-z_i)L_t(b+e_i,z)=z_jL_t(b,z)-L_{t+1}(b,z),\qquad j=3-i,\quad i=1,2,$$

$$(3.10)\quad \begin{aligned}&(c+t)L_{t+1}(b,z)-\sum_{i=1}^{2}(b_i+t)z_iL_t(b,z)+tz_1z_2L_{t-1}(b,z)\\&=-R_{t+1}(b,z)+(z_1+z_2)R_t(b,z)-z_1z_2R_{t-1}(b,z)\\&=\frac{b_1b_2}{c(c+1)}(z_1-z_2)^2R_{t-1}(b+e_1+e_2,z)\\&=\frac{(z_1-z_2)^2}{t(t+1)}D_1D_2R_{t+1}(b,z).\end{aligned}$$

In (3.10), where $c=b_1+b_2$, the second and third equalities come from two applications of [3, (5.9-24)] and [3, (5.9-9)], respectively. The case with t a nonnegative integer and $b=(1/2,1/2)$ was quoted in [6, (2.24)], where it was used to show that the quantity $\lambda_n(1/2,1/2;x,y)$ defined by [6, (1.13)] is a homogeneous polynomial of degree $2n$ in $x^{1/2}$ and $y^{1/2}$ containing $(x^{1/2}-y^{1/2})^2$ as a factor. That conclusion is confirmed by (6.13) of the present paper, which yields the formula

$$(3.11)\quad \begin{aligned}\lambda_n(1/2,1/2;x,y)=2\sum_{s=1}^{n}\binom{n}{s}\binom{n+s}{s}[\psi(n+1+s)-\psi(n+1)]\\ \cdot\left(\frac{x^{1/2}-y^{1/2}}{2}\right)^{2s}(xy)^{(n-s)/2}.\end{aligned}$$

4. The case of a zero argument. If all components of z except one, say z_k, are fixed, both R_t and L_t have a branch point at $z_k=0$. (In the exceptional case where b_k is a nonpositive integer, L_t is a polynomial in z_k, for b_k can be raised to 0 by successive applications of (3.4) and then omitted by (2.3). Equation (5.17) is an example. Similar remarks apply to R_t.) If $\operatorname{Re}(b_1+\cdots+b_{k-1}+t)>0$, R_t and L_t have finite limits as $z_k\to 0$ in $\mathbb{C}_0$ with $|\arg z_k|$ bounded away from π, and we define the value of the function at $z_k=0$ to be this limit. By an extension [3, (8.3-4)] of Gauss' theorem for a hypergeometric function with unit argument,

$$(4.1)\quad \begin{aligned}&R_t(b_1,\cdots,b_k;z_1,\cdots,z_{k-1},0)\\&=\frac{\Gamma(c)\Gamma(c-b_k+t)}{\Gamma(c+t)\Gamma(c-b_k)}R_t(b_1,\cdots,b_{k-1};z_1,\cdots,z_{k-1}),\end{aligned}$$

where $c=b_1+\cdots+b_k\neq 0,-1,-2,\cdots$; $\operatorname{Re}(c-b_k+t)>0$; and $z_1,\cdots,z_{k-1}\in\mathbb{C}_0$. Logarithmic differentiation with respect to t shows that

$$\begin{aligned}\frac{L_t(b_1,\cdots,b_k;z_1,\cdots,z_{k-1},0)}{R_t(b_1,\cdots,b_k;z_1,\cdots,z_{k-1},0)}\\=\psi(c-b_k+t)-\psi(c+t)+\frac{L_t(b_1,\cdots,b_{k-1};z_1,\cdots,z_{k-1})}{R_t(b_1,\cdots,b_{k-1};z_1,\cdots,z_{k-1})},\end{aligned}\tag{4.2}$$

with the further conditions that $c+t$, $c-b_k\neq 0,-1,-2,\cdots$. Although (4.2) will be used in the remark following Theorem 7.1, a better form for other purposes is obtained by ordinary differentiation of (4.1):

$$\begin{aligned}&L_t(b_1,\cdots,b_k;z_1,\cdots,z_{k-1},0)\\&\quad=\frac{\Gamma(c)\Gamma(c-b_k+t)}{\Gamma(c+t)\Gamma(c-b_k)}\{L_t(b_1,\cdots,b_{k-1};z_1,\cdots,z_{k-1})\\&\qquad+[\psi(c-b_k+t)-\psi(c+t)]R_t(b_1,\cdots,b_{k-1};z_1,\cdots,z_{k-1})\},\end{aligned}\tag{4.3}$$

with the same conditions of validity as for (4.1). If $k=2$ this reduces to

$$L_t(\beta,\gamma-\beta;x,0)=\frac{\Gamma(\gamma)\Gamma(\beta+t)}{\Gamma(\gamma+t)\Gamma(\beta)}x^t[\psi(\beta+t)-\psi(\gamma+t)+\log x],\tag{4.4}$$

where β, γ, $\gamma+t\neq 0$, -1, $-2,\cdots$ and $\operatorname{Re}(\beta+t)>0$. The elliptic integral (1.9) is equivalent to Legendre's first integral if $w=\infty$, and its series expansion then contains the special case of (4.4) with $\beta=\frac{1}{2}$, $\gamma=1$, and t a nonnegative integer. This case was quoted in [6, (1.10)], with the difference of ψ-functions given by [6, (1.11)] or [3, (8.3-15)]. Other special cases of (4.4) are (1.5) and

$$L_0(1,m;x,0)=\log x-(1+1/2+1/3+\cdots+1/m),\tag{4.5}$$

where m is a positive integer. If $x=m$, this tends to the negative of the Euler-Mascheroni constant as $m\to\infty$.

5. Series expansions. Let all parameters and variables be complex, define $1-z=(1-z_1,\cdots,1-z_k)$ and $|1-z|=\max_i|1-z_i|$, and assume $|1-z|<1$ and $b_1+\cdots+b_k\neq 0$, -1, $-2,\cdots$. The series expansion [3, (5.9-4)]

$$R_t(b,z)=\sum_{s=0}^{\infty}\frac{(-t)_s}{s!}R_s(b,1-z)\tag{5.1}$$

is the Dirichlet average of the binomial series

$$x^t=\sum_{s=0}^{\infty}\frac{(-t)_s}{s!}(1-x)^s,\qquad |1-x|<1,\tag{5.2}$$

where $(-t)_0=1$ and $(-t)_s=(-t)(-t+1)(-t+2)\cdots(-t+s-1)$, $s=1,2,3,\cdots$. Differentiation with respect to t is essentially the same for both series. By [3, proof of Cor. 6.3-4] both converge uniformly for $|t|<T$, where T may be arbitrarily large, and so the series may be differentiated term by term. Since $(-t)_0=1$ we assume s to be a positive integer and find

$$\begin{aligned}(d/dt)(-t)_s&=(-1)(-t+1)(-t+2)\cdots(-t+s-1)\\&\quad+(-t)(-1)(-t+2)\cdots(-t+s-1)+\cdots\\&\quad+(-t)(-t+1)\cdots(-t+s-2)(-1).\end{aligned}\tag{5.3}$$

If $(-t)_s \neq 0$ we can rewrite this as

$$(d/dt)(-t)_s = (-t)_s\left(\frac{1}{t}+\frac{1}{t-1}+\cdots+\frac{1}{t+1-s}\right) \tag{5.4}$$
$$= (-t)_s[\psi(t+1)-\psi(t+1-s)].$$

If $t=n$ where $n=0, 1, \cdots, s-1$, then (5.4) is indeterminate, but all terms on the right side of (5.3) vanish except one, leaving

$$[(d/dt)(-t)_s]_{t=n} = (-1)^{n+1} n!(s-n-1)!, \qquad n=0, 1, \cdots, s-1. \tag{5.5}$$

Therefore, if $t \neq 0, 1, 2, \cdots$, differentiation of (5.1) yields

$$L_t(b, z) = \sum_{s=1}^{\infty} \frac{(-t)_s}{s!}[\psi(t+1)-\psi(t+1-s)]R_s(b, 1-z), \qquad |1-z|<1. \tag{5.6}$$

On the other hand, if $n=0, 1, 2, \cdots$, we find

$$L_n(b, z) = \sum_{s=1}^{n} \frac{(-n)_s}{s!}[\psi(n+1)-\psi(n+1-s)]R_s(b, 1-z) \tag{5.7}$$
$$+(-1)^{n+1} n! \sum_{s=n+1}^{\infty} \frac{(s-n-1)!}{s!} R_s(b, 1-z), \qquad |1-z|<1.$$

The first sum is empty if $n=0$, whence

$$L_0(b, z) = -\sum_{s=1}^{\infty} \frac{1}{s} R_s(b, 1-z), \qquad |1-z|<1, \tag{5.8}$$

while

$$L_1(b, z) = -R_1(b, 1-z)+\sum_{s=2}^{\infty} \frac{1}{s(s-1)} R_s(b, 1-z) \tag{5.9}$$
$$= L_0(b, z)+\sum_{s=2}^{\infty} \frac{1}{s-1} R_s(b, 1-z), \qquad |1-z|<1.$$

The second equality in (5.9) is generalized by

$$\sum_{s=1}^{\infty} \frac{1}{s} R_{N+s}(b, z) = \sum_{n=0}^{N} (-1)^{n+1}\binom{N}{n} L_n(b, 1-z), \tag{5.10}$$

where $|z|<1$ and $N=0, 1, 2, \cdots$. This equation is proved by taking the Dirichlet average of the case in which all components of z are equal.

If all components of z are equal, (5.7) reduces to

$$x^n \log x - \sum_{s=1}^{n} \frac{(-n)_s}{s!}[\psi(n+1)-\psi(n+1-s)](1-x)^s$$
$$= (-1)^{n+1} n! \sum_{s=n+1}^{\infty} \frac{(s-n-1)!}{s!}(1-x)^s, \qquad |1-x|<1, \tag{5.11}$$
$$= \frac{(x-1)^{n+1}}{n+1}\, {}_2F_1(1, 1; n+2; 1-x)$$
$$= \frac{(x-1)^{n+1}}{n+1} R_{-1}(1, n+1; x, 1), \qquad |\arg x|<\pi.$$

We shall use (5.11) to prove (5.15).

An explicit formula [3, (6.2-1)] and a recurrence relation [4, (A.6)] are available for computing the polynomials R_s in (5.6) and (5.7). Moreover we may use (2.5) to make one component of z equal to unity; then R_s has one vanishing component that can be eliminated by [3, (6.2-5), (6.2-6)]. If $k=2$ this leaves a power series in one variable:

$$L_t(\beta, \gamma-\beta; 1-x, 1)=\sum_{s=1}^{\infty} \frac{(-t)_s(\beta)_s}{(\gamma)_s s!}[\psi(t+1)-\psi(t+1-s)]x^s, \tag{5.12}$$

where $|x|<1$, $\gamma \neq 0, -1, -2, \cdots$, and $t \neq 0, 1, 2, \cdots$. From (5.7) we obtain similarly, for $n=0, 1, 2, \cdots$ and $\gamma \neq 0, -1, -2, \cdots$,

$$\begin{aligned} L_n(\beta, \gamma-\beta; 1-x, 1)&-\sum_{s=1}^{n} \frac{(-n)_s(\beta)_s}{(\gamma)_s s!}[\psi(n+1)-\psi(n+1-s)]x^s \\ &=(-1)^{n+1}n! \sum_{s=n+1}^{\infty} \frac{(s-n-1)!(\beta)_s}{(\gamma)_s s!}x^s, \qquad |x|<1, \\ &=\frac{(\beta)_{n+1}(-x)^{n+1}}{(\gamma)_{n+1}(n+1)}\,{}_3F_2(1, 1, \beta+n+1; n+2, \gamma+n+1; x), \qquad |\arg(1-x)|<\pi, \end{aligned} \tag{5.13}$$

which reduces to (5.11) if $\beta=\gamma$. An important case of (5.13) is

$$\begin{aligned} L_n(1, \gamma-1; x, 1)&-\sum_{s=1}^{n} \frac{(-n)_s}{(\gamma)_s}[\psi(n+1)-\psi(n+1-s)](1-x)^s \\ &=\frac{n!}{(\gamma)_{n+1}}(x-1)^{n+1}\,{}_2F_1(1, 1; \gamma+n+1; 1-x) \\ &=\frac{n!}{(\gamma)_{n+1}}(x-1)^{n+1}R_{-1}(1, \gamma+n; x, 1), \end{aligned} \tag{5.14}$$

where n is a nonnegative integer and $|\arg x|<\pi$. If γ is a positive integer, say $\gamma=1+m$, we can sum the ${}_2F_1$-series by (5.11) to get

$$\begin{aligned} L_n(1, m; x, 1)=\sum_{s=1}^{n} \frac{(-n)_s}{(m+1)_s}[\psi(n+1)-\psi(n+1-s)](1-x)^s+\frac{n!m!}{(n+m)!}(x-1)^{-m} \\ \cdot\left\{x^{n+m}\log x-\sum_{s=1}^{n+m}\frac{(-n-m)_s}{s!}[\psi(n+m+1)-\psi(n+m+1-s)](1-x)^s\right\}, \end{aligned} \tag{5.15}$$

where n and m are nonnegative integers, $|\arg x|<\pi$, and $x \neq 1$. If $n=0$ the first sum is empty and a change of index in the second sum gives, with the help of (2.5),

$$\begin{aligned} L_0(1, m; x, y)&=\log y+\left(\frac{x}{x-y}\right)^m \log\frac{x}{y} \\ &+\sum_{s=0}^{m-1}\binom{m}{s}[\psi(1+s)-\psi(1+m)]\left(\frac{y}{x-y}\right)^s, \end{aligned} \tag{5.16}$$

where m is a nonnegative integer, $x, y \in \mathbb{C}_0$, and $x \neq y$. This result generalizes (4.5) and will be used to prove (6.18).

The series in (5.12) terminates if β is a nonpositive integer, and the conditions of validity may then be relaxed:

$$L_t(-n, \gamma+n; x, 1)=\sum_{s=1}^{n} \frac{(-n)_s(-t)_s}{(\gamma)_s s!}[\psi(t+1)-\psi(t+1-s)](1-x)^s, \tag{5.17}$$

where x is unrestricted, n is a nonnegative integer, $(\gamma)_n \neq 0$, and $(-t)_n \neq 0$. The last condition is required because (5.4) was used to prove (5.12). This result will be used to prove (6.12).

6. Quadratic transformations. The R-function of two variables has exactly two independent quadratic transformations [3, (6.9-7), (6.10-1)],

$$R_{2t}(\beta, \beta; x, y) = R_t(\beta + t, 1/2 - t; A, G), \tag{6.1}$$

$$R_t(\beta, \beta; x^2, y^2) = R_t(2\beta + t, 1/2 - \beta - t; A, G), \tag{6.2}$$

where A and G denote the squared arithmetic and geometric means of x and y,

$$A = \left(\frac{x+y}{2}\right)^2, \qquad G = xy, \tag{6.3}$$

and where x and y have positive real parts and $\beta + 1/2 \neq 0, -1, -2, \cdots$. When the right-hand sides of (6.1) and (6.2) are differentiated with respect to t, one term arises from the subscript and a second term from the t-dependence of the b-parameters. Using (1.2) and (2.12) we find, with the same conditions of validity,

$$2L_{2t}(\beta, \beta; x, y) = L_t(\beta + t, 1/2 - t; A, G) - G^t L_{-t-\beta}(-t, \beta + 1/2 + t; A/G, 1), \tag{6.4}$$

$$L_t(\beta, \beta; x^2, y^2) = C + D, \tag{6.5}$$

where, for convenience in applications, we give several forms for C and D that are connected by (2.5), (2.11), (6.2), and [3, (5.9-19)]:

$$\begin{aligned} C &= L_t(2\beta + t, 1/2 - \beta - t; A, G) \\ &= \log (G) R_t(\beta, \beta; x^2, y^2) + G^t L_t(2\beta + t, 1/2 - \beta - t; A/G, 1) \\ &= \log (A) R_t(\beta, \beta; x^2, y^2) \\ &\quad - A^{1/2-\beta} G^{t+\beta-1/2} L_{-t-\beta-1/2}(1/2 - \beta - t, 2\beta + t; A/G, 1) \\ &= \log (AG) R_t(\beta, \beta; x^2, y^2) \\ &\quad - A^{1/2-\beta} G^{2t+2\beta} L_{-t-\beta-1/2}(1/2 - \beta - t, 2\beta + t; A, G), \end{aligned} \tag{6.6}$$

$$\begin{aligned} D &= -G^t L_{-t-2\beta}(-t, \beta + 1/2 + t; A/G, 1) \\ &= \log (G) R_t(\beta, \beta; x^2, y^2) - G^{2t+2\beta} L_{-t-2\beta}(-t, \beta + 1/2 + t; A, G) \\ &= -\log (A) R_t(\beta, \beta; x^2, y^2) + A^{1/2-\beta} L_{t+\beta-1/2}(\beta + 1/2 + t, -t; A, G) \\ &= -\log (A/G) R_t(\beta, \beta; x^2, y^2) \\ &\quad + A^{1/2-\beta} G^{t+\beta-1/2} L_{t+\beta-1/2}(\beta + 1/2 + t, -t; A/G, 1). \end{aligned} \tag{6.7}$$

If $t = 0$ the second term in both (6.4) and (6.5) vanishes by (2.3):

$$\begin{aligned} 2L_0(\beta, \beta; x, y) &= L_0(\beta, 1/2; A, G), \\ L_0(\beta, \beta; x^2, y^2) &= L_0(2\beta, 1/2 - \beta; A, G), \\ L_0(1/2, 1/2; x^2, y^2) &= \log A = 2 \log \frac{x+y}{2}. \end{aligned} \tag{6.8}$$

Putting $t = -\beta$ and using the first form of C and the second form of D, we find

$$L_{-\beta}(\beta, \beta; x^2, y^2) = \log (xy) R_{-\beta}(\beta, \beta; x^2, y^2). \tag{6.9}$$

If $t=1/2-\beta$ the second term in the third form of C vanishes by (2.3), and the third form of D then gives

$$L_{1/2-\beta}(\beta,\beta;x^2,y^2)=\left(\frac{x+y}{2}\right)^{1-2\beta}L_0\left(1,\beta-1/2;\left(\frac{x+y}{2}\right)^2,xy\right). \tag{6.10}$$

From $L_{1/2-\beta}$ we can obtain $L_{-1/2-\beta}$ in terms of L_0 by (2.11) and [3, Ex. 6.10-12]. We shall use (6.10) to prove (6.18).

From the third form of C and the first form of D, we find

$$\begin{aligned}L_n(1/2+m,1/2+m;x^2,y^2)&=\log(A)R_n(1/2+m,1/2+m;x^2,y^2)\\&\quad-A^{-m}G^{n+m}L_{-n-m-1}(-n-m,n+2m+1;A/G,1)\\&\quad-G^nL_{-n-2m-1}(-n,n+m+1;A/G,1).\end{aligned} \tag{6.11}$$

If m and n are nonnegative integers, the last two terms (which are equal if $m=0$) have terminating series expansions (5.17). The result is

$$\begin{aligned}&L_n(1/2+m,1/2+m;x^2,y^2)\\&\quad=2\log\left(\frac{x+y}{2}\right)R_n(1/2+m,1/2+m;x^2,y^2)\\&\quad+\left(\frac{x+y}{2}\right)^{-2m}\sum_{s=1}^{n+m}\binom{n+m}{s}\frac{(n+m+1)_s}{(m+1)_s}\\&\quad\cdot[\psi(n+m+1+s)-\psi(n+m+1)]\left(\frac{x-y}{2}\right)^{2s}(xy)^{n+m-s}\\&\quad+\sum_{s=1}^{n}\binom{n}{s}\frac{(n+2m+1)_s}{(m+1)_s}[\psi(n+2m+1+s)-\psi(n+2m+1)]\left(\frac{x-y}{2}\right)^{2s}(xy)^{n-s}.\end{aligned} \tag{6.12}$$

By (6.2) and [3, (5.9-11)] the term in R_n can be omitted if, in the last sum, $2\log[(x+y)/2]$ is added to the terms in square brackets and the lower limit of summation is changed from 1 to 0. For example, the case $m=0$ is

$$\begin{aligned}&L_n(1/2,1/2;x^2,y^2)\\&\quad=2\sum_{s=0}^{n}\binom{n}{s}\binom{n+s}{s}\left[\psi(n+1+s)-\psi(n+1)+\log\frac{x+y}{2}\right]\left(\frac{x-y}{2}\right)^{2s}(xy)^{n-s},\end{aligned} \tag{6.13}$$

which generalizes the third equation in (6.8). This formula is useful for calculating the terms of the series (1.9). The polynomial $R_n(1/2,1/2;x^2,y^2)$ occurring in that series is the coefficient of $2\log[(x+y)/2]$ in (6.13).

Comparison of (6.12) and (1.7) gives

$$\begin{aligned}\left[\frac{\partial}{\partial\nu}P_\nu^{-m}(x)\right]_{\nu=n}&=\log\left(\frac{1+x}{2}\right)P_n^{-m}(x)\\&\quad+\left(\frac{1-x^2}{4}\right)^{m/2}\sum_{s=1}^{n-m}\frac{(m-n)_s(n+m+1)_s}{s!(m+s)!}\\&\quad\cdot[\psi(n+m+1+s)-\psi(n+m+1)]\left(\frac{1-x}{2}\right)^s\\&\quad+\left(\frac{1-x}{1+x}\right)^{m/2}\sum_{s=1}^{n}\frac{(-n)_s(n+1)_s}{s!(m+s)!}[\psi(n+1+s)-\psi(n+1)]\left(\frac{1-x}{2}\right)^s,\end{aligned} \tag{6.14}$$

where m and $n-m$ are nonnegative integers and $-1<x\leqq 1$. The logarithmic term can be incorporated in the second sum by adding $\log[(1+x)/2]$ to the terms in square brackets and beginning the sum at $s=0$. Equation (6.14) generalizes a formula due to Schelkunoff [13, (114)], [12, p. 173] for the case $m=0$, in which the two sums are equal. Schelkunoff used his formula in the theory of conical antennas. If m is an integer, differentiation of

$$\frac{P_\nu^m(x)}{\Gamma(\nu+m+1)}=(-1)^m\frac{P_\nu^{-m}(x)}{\Gamma(\nu-m+1)},\qquad -1\leqq x\leqq 1 \tag{6.15}$$

gives

$$\begin{aligned}\left[\frac{\partial}{\partial\nu}P_\nu^m(x)\right]_{\nu=n}&=(-1)^m\frac{(n+m)!}{(n-m)!}\left[\frac{\partial}{\partial\nu}P_\nu^{-m}(x)\right]_{\nu=n}\\&\quad+[\psi(n+m+1)-\psi(n-m+1)]P_n^m(x),\qquad n\geqq m.\end{aligned}\tag{6.16}$$

If m is even, there is no branch point at $x=1$, and the last three equations then hold for $|\arg(1+x)|<\pi$.

Equation (6.14) requires $n\geqq m$, but a known formula [9, 8.762(2)] for the case $n=0$, $m=1$ can be generalized by putting $\beta=1/2+m$ in (6.10) to get

$$L_{-m}(1/2+m,1/2+m;x^2,y^2)=\left(\frac{x+y}{2}\right)^{-2m}L_0\left(1,m;\left(\frac{x+y}{2}\right)^2,xy\right).\tag{6.17}$$

If m is a nonnegative integer, the L_0 function has a terminating series given by (5.16). The result is

$$\begin{aligned}&L_{-m}(1/2+m,1/2+m;x^2,y^2)\\&\quad=\left(\frac{2}{x-y}\right)^{2m}\log\frac{(x+y)^2}{4xy}+\left(\frac{2}{x+y}\right)^{2m}\\&\qquad\cdot\left\{\log(xy)+\sum_{s=0}^{m-1}\binom{m}{s}[\psi(1+s)-\psi(1+m)]\left(\frac{2}{x-y}\right)^{2s}(xy)^s\right\},\end{aligned}\tag{6.18}$$

where x and y have positive real parts, $x\neq y$, and $m=0,1,2,\cdots$. From L_{-m} we can obtain also L_{-m-1} by using (2.11) and [3, Ex. 6.10-12]. Comparison of (6.18) and (1.7) shows that

$$\begin{aligned}m!\left[\frac{\partial}{\partial\nu}P_\nu^{-m}(x)\right]_{\nu=0}&=(-1)^m\left(\frac{1+x}{1-x}\right)^{m/2}\log\frac{1+x}{2}\\&\quad+\left(\frac{1-x}{1+x}\right)^{m/2}\sum_{s=0}^{m-1}\binom{m}{s}[\psi(1+s)-\psi(1+m)]\left(\frac{2}{x-1}\right)^s,\end{aligned}\tag{6.19}$$

where $-1<x<1$ and $m=0,1,2,\cdots$. The limit as $x\to 1$ is 0. If m is even, the condition $-1<x<1$ can be replaced by $|\arg(x+1)|<\pi$ and $x\neq 1$.

The infinite series given in [9, (8.761)] can be reproduced by using (1.7), (6.5) with the third form of both C and D, and (5.12). The second equality in (5.4) is used also. It suffices to assume $\mu\neq -1,-2,-3,\cdots$ instead of $\operatorname{Re}\mu>-1$.

7. Inequalities. We assume t real and b_i and x_i strictly positive for $i=1,\cdots,k$. The largest and smallest of the x_i are denoted by $x_{\max}$ and $x_{\min}$, respectively, and we assume $x_{\max}>x_{\min}$ to exclude trivialities. Writing L_t for $L_t(b,x)$ and R_t for $R_t(b,x)$, we see at once from (1.1) and (1.3) that

$$R_t\log x_{\min}<L_t<R_t\log x_{\max}.\tag{7.1}$$

Chebyshev's inequality for integrals [10, Thm. 236] implies

$$L_t > R_t L_0, \qquad t>0, \tag{7.2}$$

with reversed inequality for $t<0$. Since $R_0=1$, (7.1) and (7.2) are subsumed in the following theorem, which is used in [6, (3.33)].

THEOREM 7.1. *Both L_t and L_t/R_t are strictly increasing functions of t. The limit of L_t/R_t is* $\log x_{\max}$ *as $t\to+\infty$ and* $\log x_{\min}$ *as $t\to-\infty$.*

Proof. Since the integrand of (1.1) is log-convex in t, R_t is strictly log-convex in t by [3, Ap. B.6] or [1, Thm. 4]. Hence its derivative and logarithmic derivative are strictly increasing, which proves the first part of the theorem. To prove the second part we use [1, Thm. 3]:

$$\lim_{t\to+\infty} (R_t)^{1/t} = x_{\max}, \tag{7.3}$$

which implies that $R_t\to\infty$ if $x_{\max}>1$ and $R_t\to 0$ if $x_{\max}<1$. If $x_{\max}=1$ then $x_i<1$ for some i (since $x_{\max}>x_{\min}$), and $u\cdot x$ in (1.1) is less than unity except on the set of measure zero where $u_i=0$. Hence $(u\cdot x)^t\to 0$ almost everywhere as $t\to+\infty$, and so $R_t\to 0$. Thus $|\log R_t|\to\infty$ in all cases, and we may use L'Hôpital's rule to conclude from (7.3) that

$$\log x_{\max} = \lim_{t\to+\infty}\frac{\log R_t}{t} = \lim_{t\to+\infty}\frac{L_t}{R_t}.$$

The proof for $t\to-\infty$ is similar.

Remark. If the assumptions of Theorem 7.1 are changed so that exactly one of the variables, say x_k, is 0, then R_t and L_t are well defined for $t>-b_1-\cdots-b_{k-1}$. In this region L_t/R_t increases strictly with t and has limit $\log x_{\max}$ as $t\to+\infty$. This follows from (4.2) and two well-known facts: $\psi''(x)<0$ for $x>0$, implying $\psi'(c-b_k+t)-\psi'(c+t)>0$; and $\psi(x)=\log x+O(1/x)$ as $x\to+\infty$, implying $\psi(c-b_k+t)-\psi(c+t)\to 0$ as $t\to+\infty$.

THEOREM 7.2. *Let t be real and $x_{\max}>x_{\min}>0$. Let $cw=(cw_1,\cdots,cw_k)$, where $c>0$ and the w's are positive weights with $\sum w_i=1$. Then the limit of $L_t(cw,x)$ is $\sum w_i x_i^t\log x_i$ as $c\to 0+$ and $(\sum w_i x_i)^t\log(\sum w_i x_i)$ as $c\to+\infty$. Also, $L_0(cw,x)$ is strictly increasing and strictly concave in c, and $L_1(cw,x)$ is strictly decreasing and strictly convex in c.*

Proof. We assume for the moment that $x_{\max}<1$, so that the series expansions (5.6) and (5.7) converge. The proof of the first part of the theorem is entirely similar to the proof of [1, Thm. 1], in which L is the L_0 of the present paper. The second part of the theorem follows from (5.8), (5.9), and [7, Thm. 5]. The case $x_{\max}\geqq 1$ reduces to the case $x_{\max}<1$ by use of (2.5) and [1, Thm. 1].

Remark. Example 5 in § 1 is contained in Theorem 7.2. The underlying reason why L_0 and L_1 are exceptionally simple is that $x^t\log x$ is concave in x for all $x>0$ only if $t=0$ and convex for all $x>0$ only if $t=1$.

8. Special cases. Several cases of L_t with restrictions on the parameters have been evaluated in finite terms in (4.6), (5.15), (5.16), (5.17), (6.12), (6.13) and (6.18). Further cases can be evaluated from these by (2.11) and the relations between associated functions in § 3. In (5.15) and (5.17) the apparent restriction of a unit variable can be removed by (2.5). We mention here some other cases, starting with

$$L_0(\beta, 1-2\beta; 1, 0) = \psi(\beta)-\psi(1-\beta) = -\pi\cot(\pi\beta), \tag{8.1}$$

where β is not an integer and $\operatorname{Re}\beta>0$, and

$$L_{2t}(\beta,\beta;x,-x)=\frac{\pi^{1/2}\Gamma(\beta+1/2)}{2\Gamma(1/2-t)\Gamma(\beta+1/2+t)}(-x^2)^t \cdot[\psi(1/2-t)-\psi(\beta+1/2+t)+\log(-x^2)], \tag{8.2}$$

where $|\arg(\pm x)|<\pi$, $|\arg(-x^2)|<\pi$, and $\beta+1/2\neq 0,-1,-2,\cdots$. Equation (8.1) comes from (1.5) or (4.5), and (8.2) comes from differentiating the formula in [3, Ex. 8.3-9].

To evaluate an R-function or L-function of two variables with positive integral b-parameters, we raise the b-parameters from unity by successive differentiations using [3, (5.9-9)]:

$$R_{t-1}(m,m';x,y)=\frac{(m+m'-1)!}{(m-1)!(m'-1)!(t)_{m+m'-2}} \cdot D_x^{m-1}D_y^{m'-1}R_{t+m+m'-3}(1,1;x,y). \tag{8.3}$$

Use of [3, Ex. 5.9-13] and two applications of Leibnitz' rule lead to

$$R_{t-1}(m,m';x,y)=(m)_{m'}x^{t-1}\sum_{s=0}^{m-1}\frac{(1-m)_s(m')_s}{(t)_{m'+s}s!}\left(\frac{x}{x-y}\right)^{m'+s} +(m')_m y^{t-1}\sum_{s=0}^{m'-1}\frac{(1-m')_s(m)_s}{(t)_{m+s}s!}\left(\frac{y}{y-x}\right)^{m+s}, \tag{8.4}$$

where m and m' are positive integers, $x,y\in\mathbb{C}_0$, $x\neq y$, and $(t)_{m+m'-1}\neq 0$. With the same conditions, differentiation with respect to t gives

$$\begin{aligned} L_{t-1}(m,m';x,y)=(m)_{m'}x^{t-1}&\sum_{s=0}^{m-1}\frac{(1-m)_s(m')_s}{(t)_{m'+s}s!} \\ &\cdot[\psi(t)-\psi(t+m'+s)+\log x]\left(\frac{x}{x-y}\right)^{m'+s} \\ +(m')_m y^{t-1}&\sum_{s=0}^{m'-1}\frac{(1-m')_s(m)_s}{(t)_{m+s}s!} \\ &\cdot[\psi(t)-\psi(t+m+s)+\log y]\left(\frac{y}{y-x}\right)^{m+s}. \end{aligned} \tag{8.5}$$

Agreement with the special case (5.16) can be shown by a binomial expansion of $[1+y/(x-y)]^m$.

Putting $n=0$ and $\gamma=1$ in (5.13), dividing by β, and letting $\beta\to 0$, we find a relation between L_0 and Euler's dilogarithm:

$$-\lim_{\beta\to 0}\beta^{-1}L_0(\beta,1-\beta;1-x,1)=x\,{}_3F_2(1,1,1;2,2;x)=\sum_{n=1}^{\infty}\frac{x^n}{n^2}. \tag{8.6}$$

By (2.3) the limit of the same quantity as β tends to 1 is

$$-\lim_{\beta\to 1}\beta^{-1}L_0(\beta,1-\beta;1-x,1)=-\log(1-x)=\sum_{n=1}^{\infty}\frac{x^n}{n}. \tag{8.7}$$

The conditions of validity for (8.5) exclude the function

$$L_{-1}(1,1;x,y)=\frac{(\log x)^2-(\log y)^2}{2(x-y)},\qquad x\neq y, \tag{8.8}$$

which is evaluated by putting $f(x) = (\log x)^2$ in [3, (5.5-3)]. Similarly putting $f(x) = x^t$ in [3, Ex. 5.5-1], we find

$$R_{t-2}(1, 1, 1; x, y, z) = \frac{2}{t(t-1)} \sum_{\text{cyc}} \frac{x^t}{(x-y)(x-z)}, \tag{8.9}$$

where $t \neq 0$, 1 and the sum extends over cyclic permutations of the distinct complex numbers x, y, z. We shall refrain from raising the b-parameters by differentiation with respect to x, y, z as in (8.3). Differentiation with respect to t gives

$$L_{t-2}(1, 1, 1; x, y, z) = \frac{2}{t(t-1)} \sum_{\text{cyc}} \frac{x^t}{(x-y)(x-z)} \left(\log x - \frac{1}{t} - \frac{1}{t-1} \right), \tag{8.10}$$

with the same conditions of validity as for (8.9). This is the function that occurs in (1.8). The exceptional cases with $t = 0$ or 1 are evaluated by successively putting $f(x) = \log x$, $(\log x)^2$, $x \log x$, and $x(\log x)^2$ in [3, Ex. 5.5-1]:

$$R_{-2}(1, 1, 1; x, y, z) = -2 \sum_{\text{cyc}} \frac{\log x}{(x-y)(x-z)}, \tag{8.11}$$

$$L_{-2}(1, 1, 1; x, y, z) = -\sum_{\text{cyc}} \frac{(\log x)^2 + 2 \log x}{(x-y)(x-z)}, \tag{8.12}$$

$$R_{-1}(1, 1, 1; x, y, z) = 2 \sum_{\text{cyc}} \frac{x \log x}{(x-y)(x-z)}, \tag{8.13}$$

$$L_{-1}(1, 1, 1; x, y, z) = \sum_{\text{cyc}} \frac{x[(\log x)^2 - 2 \log x]}{(x-y)(x-z)}. \tag{8.14}$$

Values for special cases of the L-function permit evaluation of some special cases of ${}_3F_2$ by using (5.13). For example,

$${}_3F_2(1, 1, m+n+3/2; n+2, 2m+n+2; x)$$

can be found from (5.13) and (6.12) if m and n are nonnegative integers. The case $n = 0$ is

$$\begin{aligned} {}_3F_2(1, 1, m+3/2; 2, 2m+2; x) &= (-4/x) \log y - (2/x) y^{-2m} \sum_{s=1}^{m} \binom{m}{s} \\ &\quad \cdot [\psi(m+1+s) - \psi(m+1)](y-1)^{2s}(1-x)^{(m-s)/2}, \end{aligned} \tag{8.15}$$

where $2y = 1 + (1-x)^{1/2}$, $|\arg(1-x)| < \pi$, and $m = 0, 1, 2, \cdots$.

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